# Stanford PAI

Section 1

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USA 

India 

Greece 

Italy 

Czech Republic 

Belgium 

Spain 

Senegal 

# Hello!

Please keep webcams on

A few words about yourself

… I will call in order based on my screen

# Review: **Event**

Some **outcome**, or set of **outcomes**, of interest in a setting

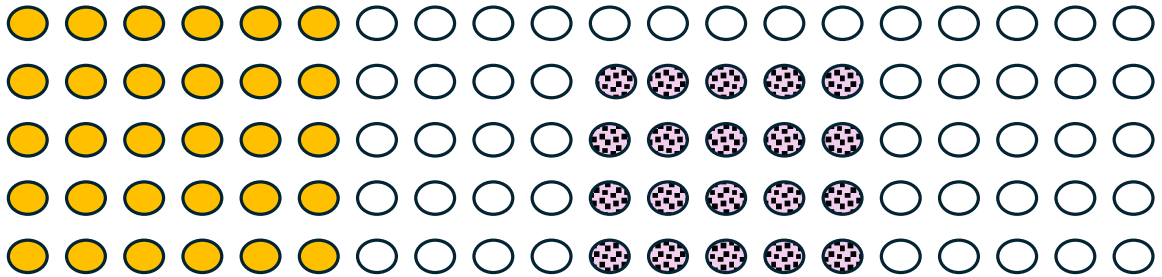
| **Setting** | **Possible events** |
|---|---|
| Single coin toss | Heads, Tails |
| Person comes to store | Buys A, Buys A and B … |
| Two coin tosses | HH, HT, … |

# We Compute Probabilities of **Events**

| **Setting** | **Possible events** | **Example** |
|---|---|---|
| Single coin toss | H, T | P(H) = 0.5 |
| Person comes to store | Buys A, Buys A and B … | P(A) = 0.05<br>P(A & B) = 0.001 |
| Two coin tosses | HH, HT, … | P(HH) = 0.25<br>P(One T) = 0.5 |

# Visualizing Probabilities

100 days



**R**ained on 30 days

**S**nowed on 20 days

**C**lear on 50 days

Never rained **and** snowed on any day

**Pick a random day**

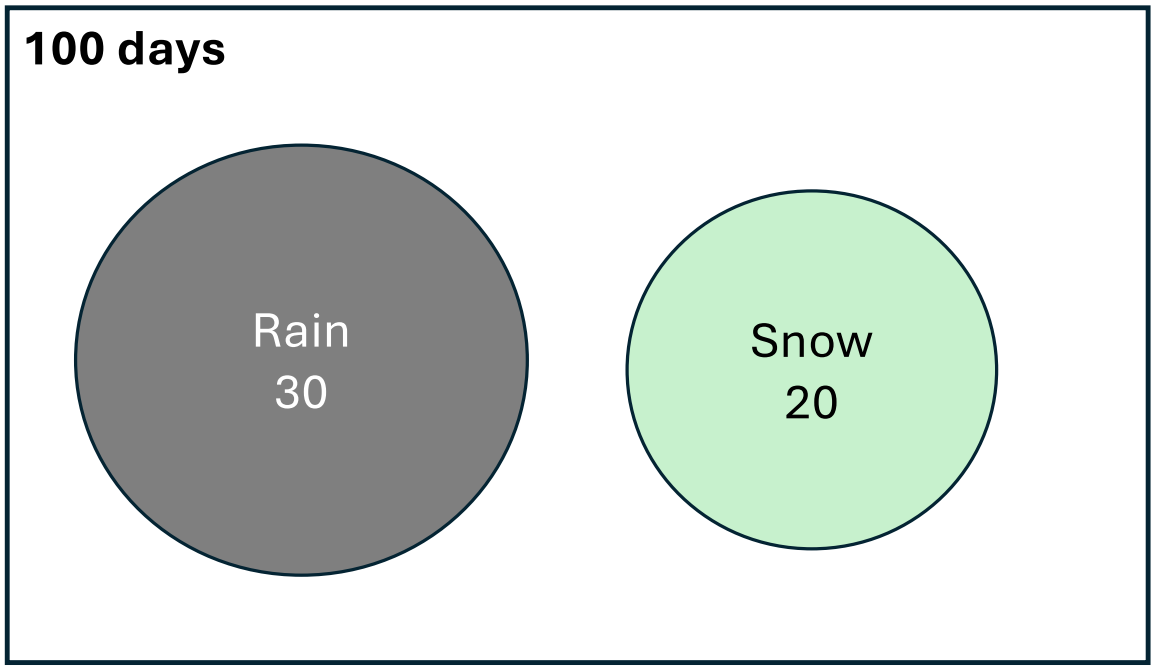
P(R) = 0.3

P(S) = 0.2

P(R and S) = 0

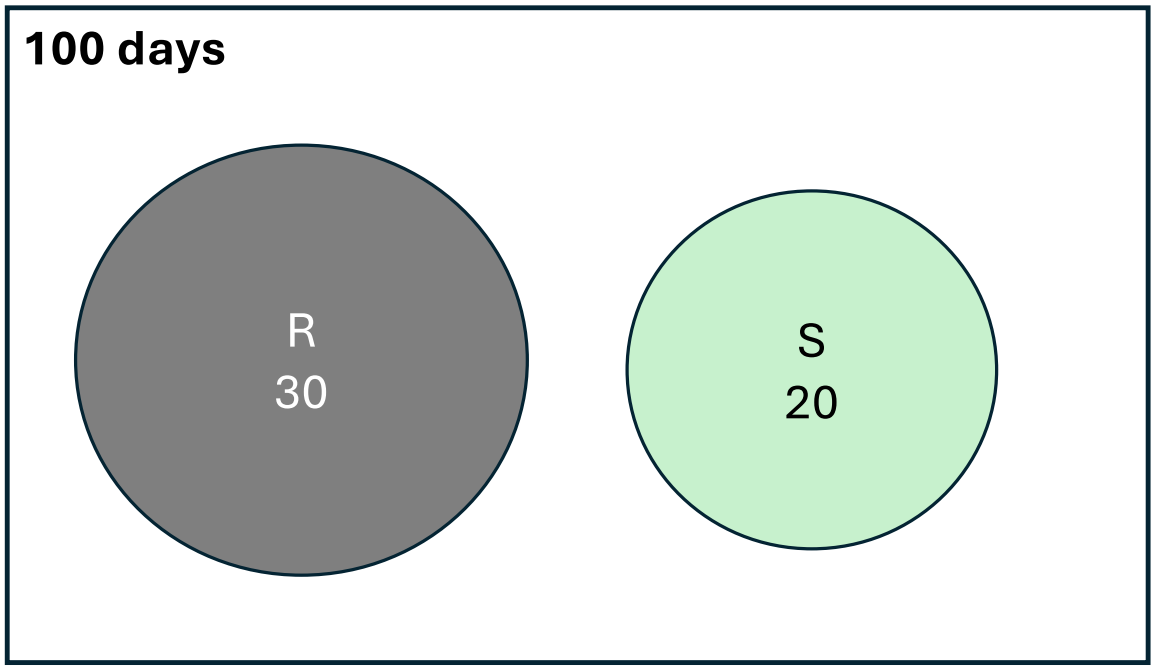
P(C) = 0.5

# Venn Diagram Representation of Probability (P)




**R and S are mutually exclusive**

# Simple Examples

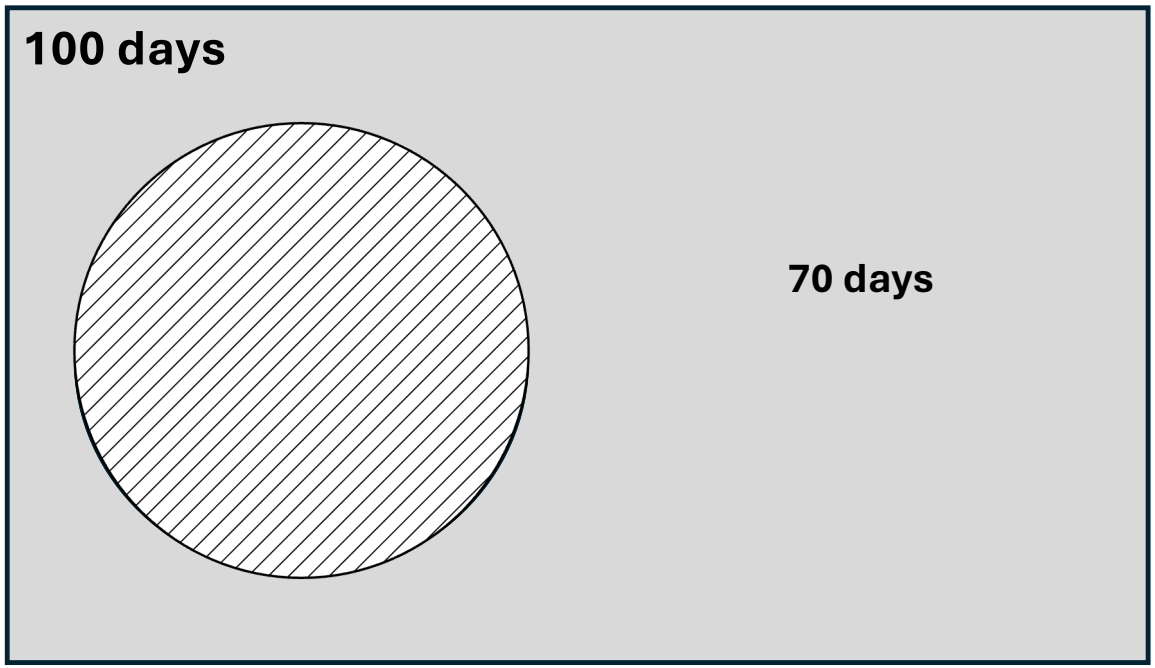



P(R) = 0.3
P(S) = 0.2

P(R or S) = 0.5

P(R and S) = 0: cannot happen in our scenario

# Complement of an Event




Complement of event R is: R does not happen -- $R^C$

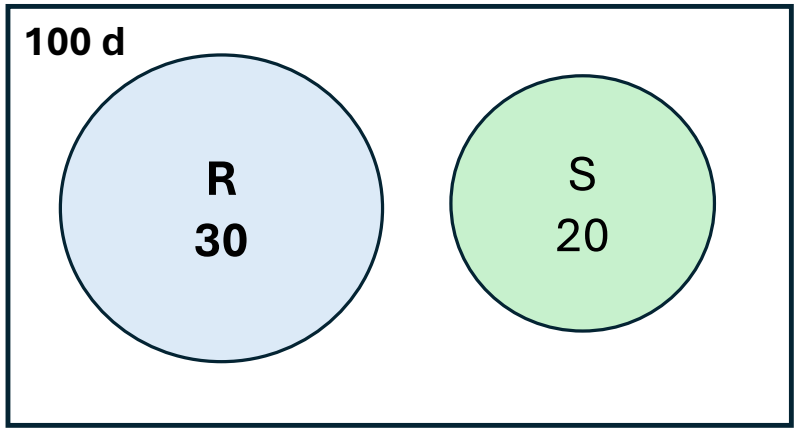
70 no rain days

$P(R^C) = 1 - P(R)$

$P(R^c) = 1 - 0.3 = 0.7$

# P (R or S)

## Mutually Exclusive

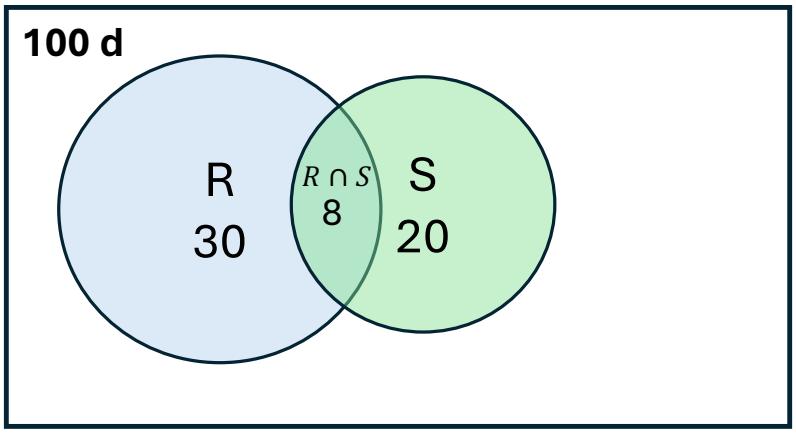



P(R or S) = 0.5: 50 days

**Mutially exclusive**

P(R or S) = P(R) + P(S)

## Not Mutually Exclusive



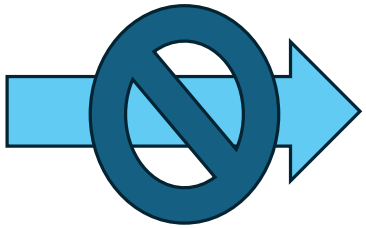

8 days of both R and S: Simply adding double-counts overlapped days $R \cap S$

**Not mutually exclusive**

$$P(R \ or \ S) = P(R) + P(S) - P(R \cap S)$$

# Independence of Events -- Caution

Outcome of one coin toss

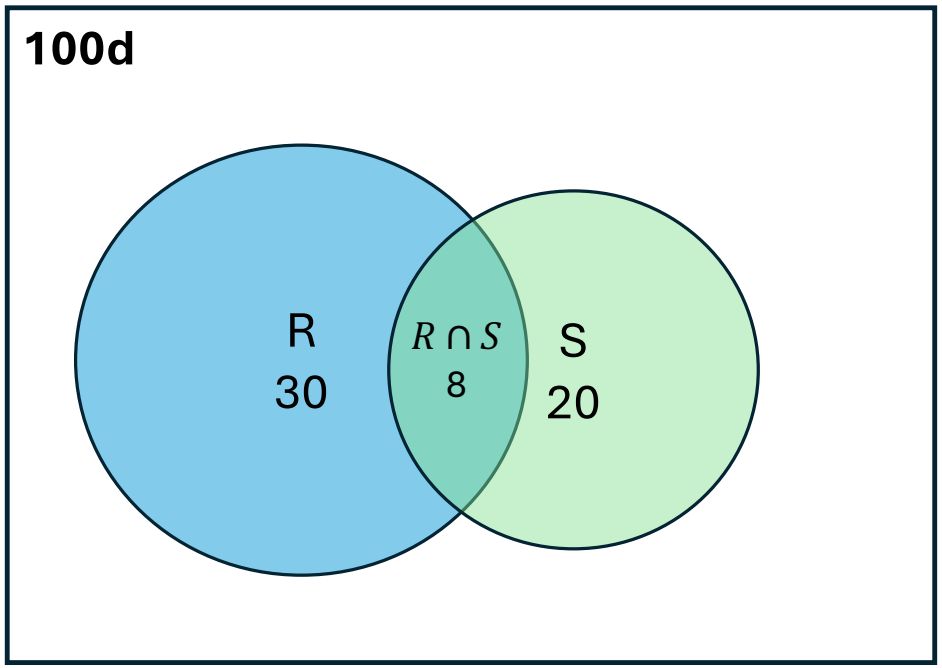


Outcome of another coin toss

Independence sometimes thought of as one event having no "influence" on another

Sometimes correct, but misses the real concept

# Effect of Knowing About an Event



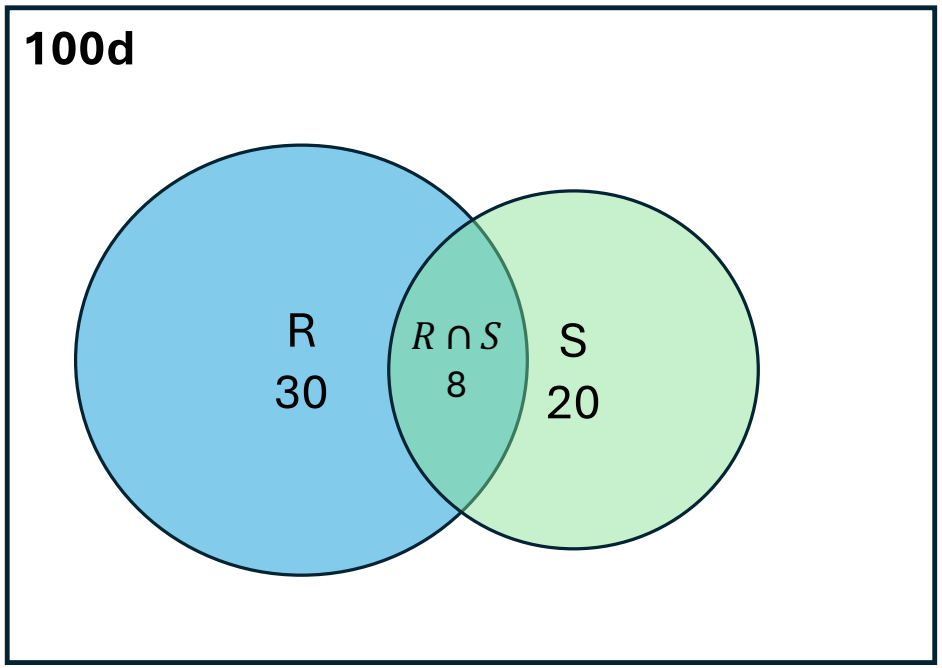

Not to scale!

P(R) = 0.3

R ∩ $S$ is the same as "R and S"

Does knowing that it snowed add any information about whether it rained?

# Unconditional Vs Conditional Probability



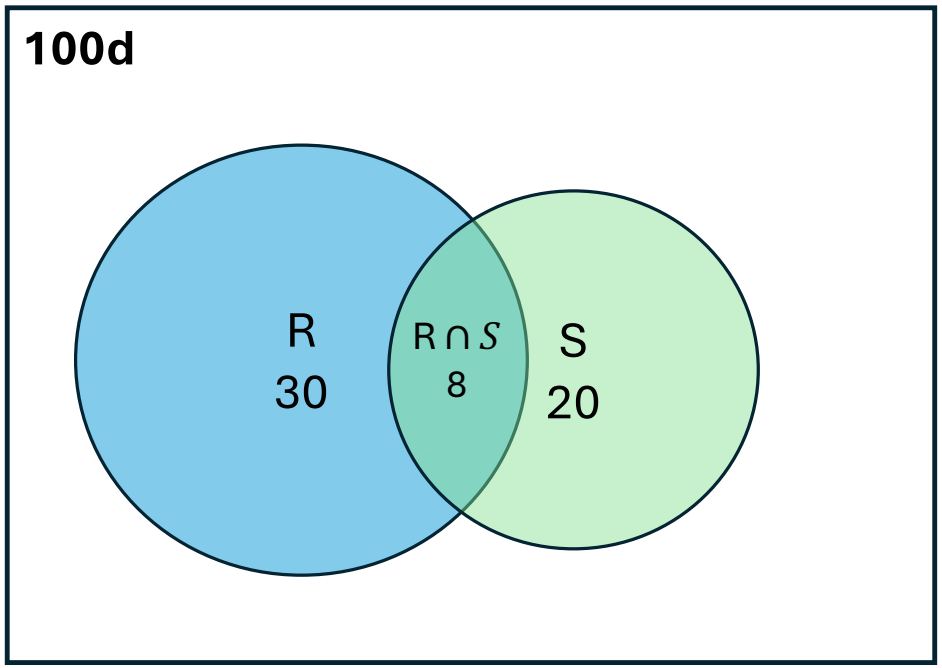

Not to scale!

P(R) = 0.3 (unconditional)

P(R|S): probability it rained, given that it has snowed

**Conditional Probability**

# Computing Conditional Probability



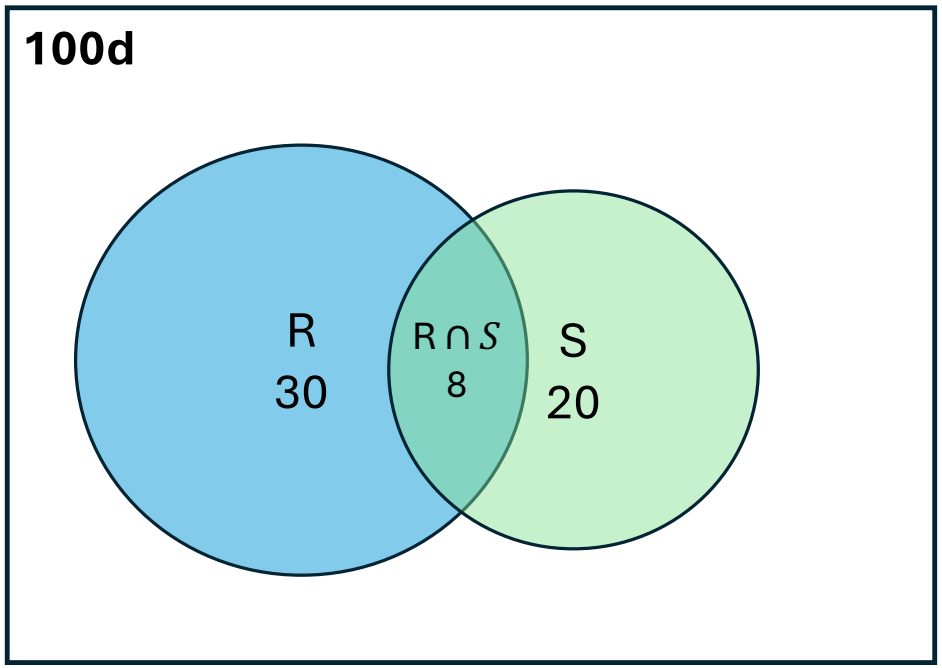

Not to scale!

Given a snow day, our universe is only snow days

Out of the 20 snow days, 8 days also had rain

P(R|S) = 8/20 = 0.4

# Computing Conditional Probability

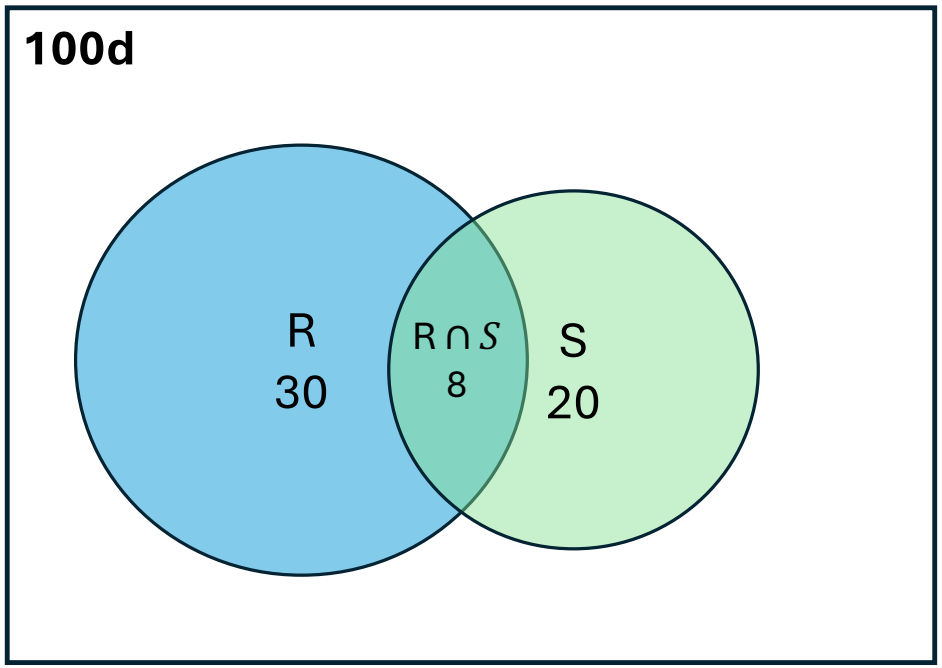



Not to scale!

If you pick a random snow day, the probability it had also rained is 8/20

$P(R|S) = 8/20 = 0.4$

# Independence is Only About Information




Not to scale!

$$P(R|S) = \frac{P(R \cap S)}{P(S)} = 0.4$$

$$P(S|R) = \frac{P(R \cap S)}{P(R)} = 0.2667$$

$P(R|S) \neq P(R)$; $P(S|R) \neq P(S)$

Knowing that one occurred changed the probability of the other: **Not independent**

# Independence: Uncond P = Conditional P

**Events A and B are independent if and only if:**

$P(A|B) = P(A)$

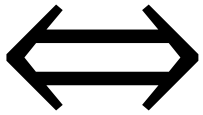
Which also means that

$P(B|A) = P(B)$

Knowing that one event occurred gives no **information** about the other

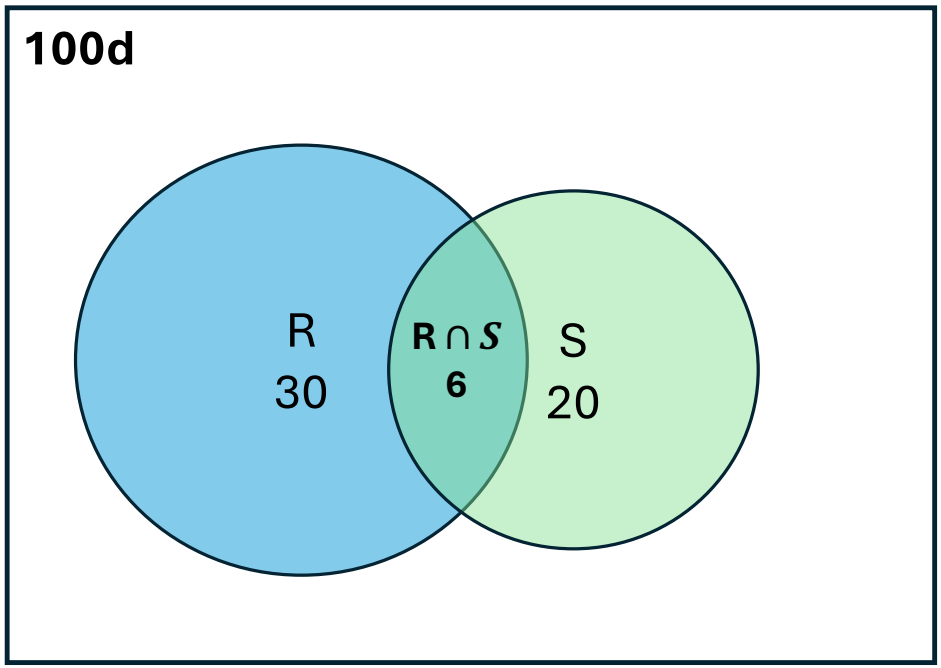
# The Crux of Independence

Unconditional and conditional probabilities of two events are equal



**Independent**

# Change Level of  Uncertainty or Not?



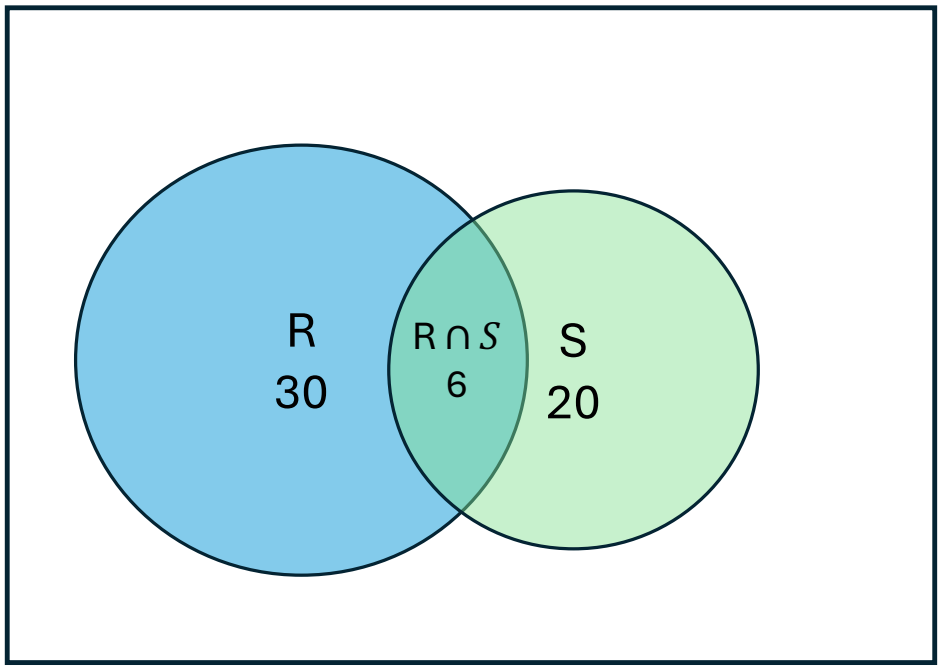

Not to scale!

$$P(R|S) = \frac{P(R \cap S)}{P(S)} = 0.3 = P(R)$$

$$P(S|R) = \frac{P(R \cap S)}{P(R)} = 0.2 = P(S)$$

**Independent**: knowing that one occurred **did not** change the probability of the other

# Independent? Multiply




Not to scale!

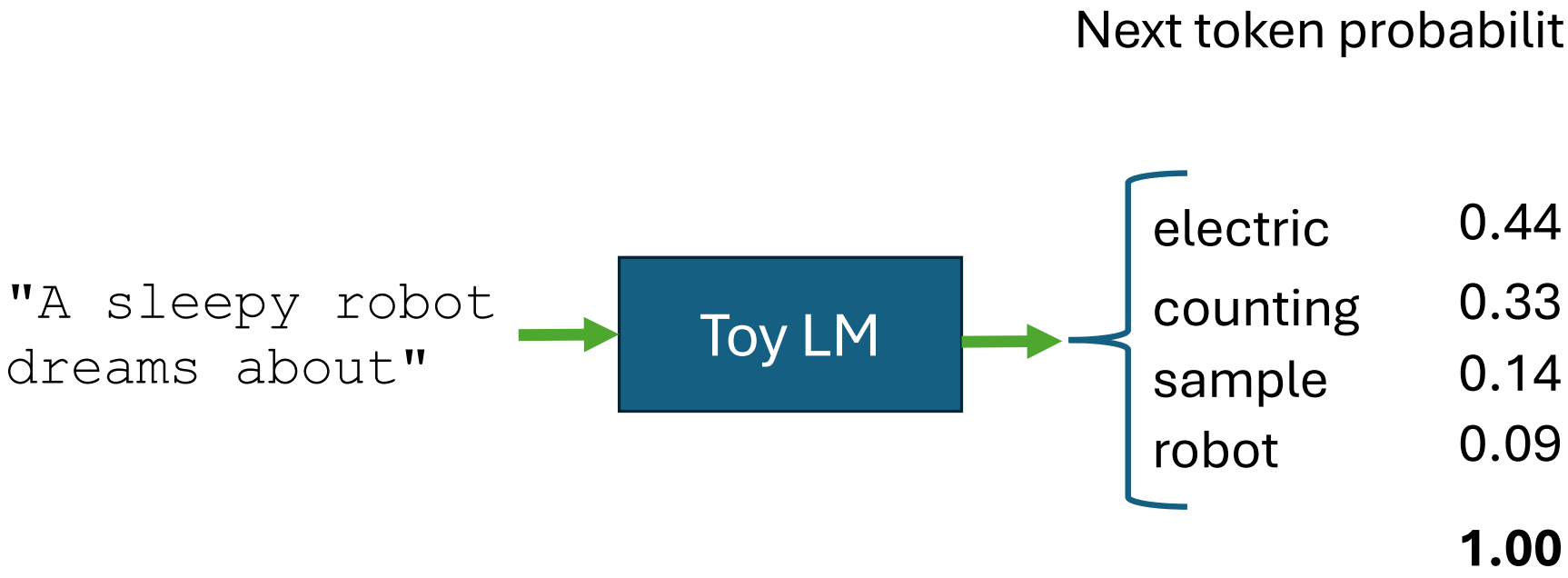
R and S independent:

P(R and S) = P(R) * P(S)

# Problem of the Week

*If I give the exact same prompt to an LLM twice, should I expect the same result? Is getting a different result a feature or a bug?*

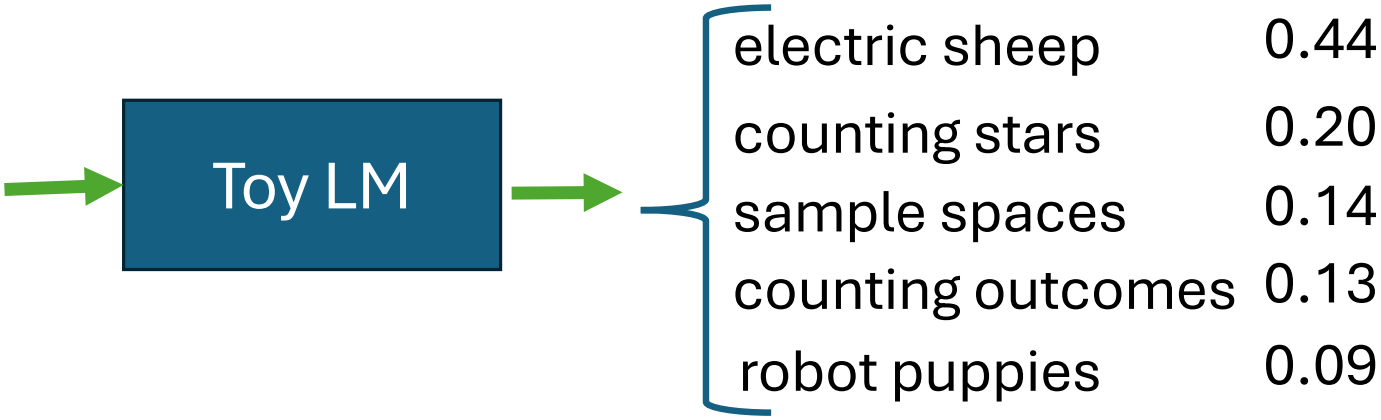
# Toy LM – first token Probabilities




The LLM will then take its output, add it to the input and generate the next token and so on until **completion**.

# Toy LLM – **All completions**

"A sleepy robot dreams about"




Probability of getting different completions on two different tries.

# Two Tries – How Many Possibilities?

| First try | Second try | Probability |
|---|---|---|
| electric sheep | counting stars | 0.44*0.2 |
| sample spaces | electric sheep | 0.14*0.44 |
| robot puppies | counting outcomes | 0.09*0.13 |
| counting stars | counting stars | 0.13*0.13 |
| sample spaces | robot puppies | 0.14*0.09 |
| electric sheep | robot puppies | 0.44*0.09 |
| … | … | … |

**25**

**possibilities**

# Event: Getting Different Results

| First try | Second try | Probability |
|---|---|---|
| electric sheep | counting stars | 0.44*0.2 |
| sample spaces | electric sheep | 0.14*0.44 |
| robot puppies | counting outcomes | 0.09*0.13 |
| counting stars | counting stars | 0.13*0.13 |
| sample spaces | robot puppies | 0.14*0.09 |
| electric sheep | robot puppies | 0.44*0.09 |
| … | … | … |

Correct, but we can do it more easily

Compute and add up all the probabilities for the cases that have two different results (20 cases)

# Easier: P(Same completion)$^C$

Far fewer events to count – only 5

P(counting stars, counting stars) = 0.20*0.20 = $0.20^2$

P( robot puppies, robot puppies) = 0.09 * 0.09 = $0.09^2$

P(outcome i, outcome i) = $p_i$ * $p_i$ = $p_i^2$

Same on both tries: "electric sheep" twice **or**
"counting stars" twice robot puppies twice**, or …**

P(different result) = 1 – P(same result)

"electric sheep" twice ($p=0.44^2$) **or** "counting stars" twice ($p=0.2^2$) **or** ...

***all mutually exclusive***

P(Same on both) = $p_1^2 + p_2^2 + p_3^2 + p_4^2 + p_5^2$

$= \sum_{i=1}^{5} p_i^2$

$\sum$ is math shorthand for "sum"

P(different on two tries) = $1 - \sum_{i=1}^{5} p_i^2 \approx 0.72$

# Optional Problem

*Each call says "robot puppies" with probability 0.09. If we prompt twice, what is the probability that "robot puppies" appears at least once?*

1 – P("robot puppies" does not occur at all)

$R_i$: "robot puppies" occurs in trial i

$R_i^c$: "robot puppies" **does not** occur on trial i

$P(R_i^c)$ = 0.91 for i = 1,2

# Optional Problem

*Each call says "robot puppies" with probability 0.09. If we prompt twice, what is the probability that "robot puppies" appears at least once?*

$P(R_i^c)$ = 0.91 for i = 1,2

$P(none)$ = $P(R_1^c and R_2^c) = 0.91 * 0.91 = 0.83$

$P(at\ least\ one)$ = $P(none^c)$ = $1 - P(none)$

$= 1 - 0.83$ = 0.17